NFL Referee Penalties: Data Loading and EDA

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2026-04-05

- use `nflreadr::load_pbp()` for penalty events and game context,
- use `nflreadr::load_officials()` for game-level official assignments,

```
seasons <- 2015:2025
season_type_keep <- c("REG")
```

1) Load datasets

```
pbp_raw <- load_pbp(seasons) %>%
  clean_names()

pbp_raw %>% dim()
```

```
## [1] 532376    372
```

```
officials_raw <- load_officials(seasons) %>%
  clean_names()

officials_raw %>% dim()
```

```
## [1] 21900     9
```

```
head(officials_raw)
```

```
## # A tibble: 6 x 9
##   game_id    game_key official_name  position   jersey_number official_id season
##   <chr>      <chr>    <chr>          <chr>              <int> <chr>        <int>
## 1 2015091000 56503    Brad Freeman   Field Jud~            88 25            2015
## 2 2015091000 56503    Kent Payne     Head Line~            79 28            2015
## 3 2015091000 56503    Terrence Miles Back Judge           111 139           2015
## 4 2015091000 56503    Tim Podraza    Line Judge            47 123           2015
## 5 2015091000 56503    Scott Novak    Side Judge             1 94            2015
## 6 2015091000 56503    Undrey Wash    Umpire                96 134           2015
## # i 2 more variables: season_type <chr>, week <int>
```

2) EDA: Play-by-play data

```
pbp <- pbp_raw %>%
  filter(season %in% seasons) %>%
  filter(if ("season_type" %in% names(.)) season_type %in% season_type_keep else TRUE) %>%
  filter(!is.na(game_id))

pbp %>% summarise(rows = n(), games = n_distinct(game_id), seasons = n_distinct(season))
```

```
## # A tibble: 1 x 3
##     rows games seasons
##    <int> <int>   <int>
## 1 508914  2895      11
```

```
pbp %>%
  select(c("game_id", "season", "week", "home_team", "away_team", "posteam", "defteam",
                "qtr", "down", "play_type", "desc", "penalty", "penalty_yards")) %>%
  slice_head(n = 5)
```

```
## # A tibble: 5 x 13
##   game_id season  week home_team away_team posteam defteam   qtr  down play_type
##   <chr>    <int> <int> <chr>     <chr>     <chr>   <chr>   <dbl> <dbl> <chr>
## 1 2015_0~   2015     1 DEN       BAL       <NA>    <NA>        1    NA <NA>
## 2 2015_0~   2015     1 DEN       BAL       BAL     DEN         1    NA kickoff
## 3 2015_0~   2015     1 DEN       BAL       BAL     DEN         1     1 pass
## 4 2015_0~   2015     1 DEN       BAL       BAL     DEN         1     2 run
## 5 2015_0~   2015     1 DEN       BAL       BAL     DEN         1     3 pass
## # i 3 more variables: desc <chr>, penalty <dbl>, penalty_yards <dbl>
```

Identify penalty plays

```
penalty_plays <- pbp %>%
  filter(if ("penalty" %in% names(.)) penalty == 1 else FALSE)

penalty_plays %>%
  summarise(
    penalty_rows = n(),
    penalty_games = n_distinct(game_id),
    pct_of_all_rows = n() / nrow(pbp)
  )
```

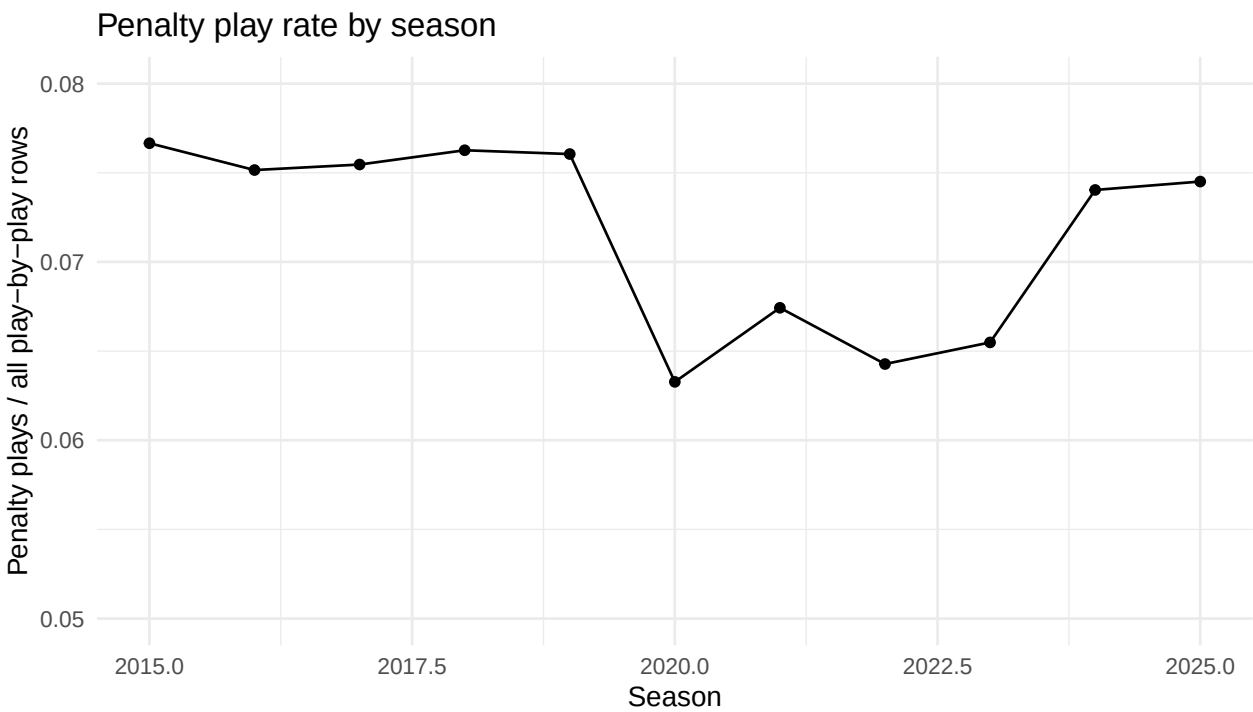
```
## # A tibble: 1 x 3
##   penalty_rows penalty_games pct_of_all_rows
##          <int>         <int>           <dbl>
## 1        36459          2895          0.0716
```

season	plays	penalties	penalty_rate	avg_penalty_yards
2015	46141	3537	7.67%	8.49
2016	45707	3435	7.52%	8.61
2017	45268	3416	7.55%	8.65
2018	45120	3441	7.63%	8.55
2019	45339	3448	7.60%	8.49
2020	45406	2873	6.33%	8.66
2021	47651	3213	6.74%	8.62
2022	47157	3031	6.43%	8.24
2023	47399	3104	6.55%	8.24
2024	47274	3500	7.40%	8.10
2025	46452	3461	7.45%	8.03

Penalty rate by season

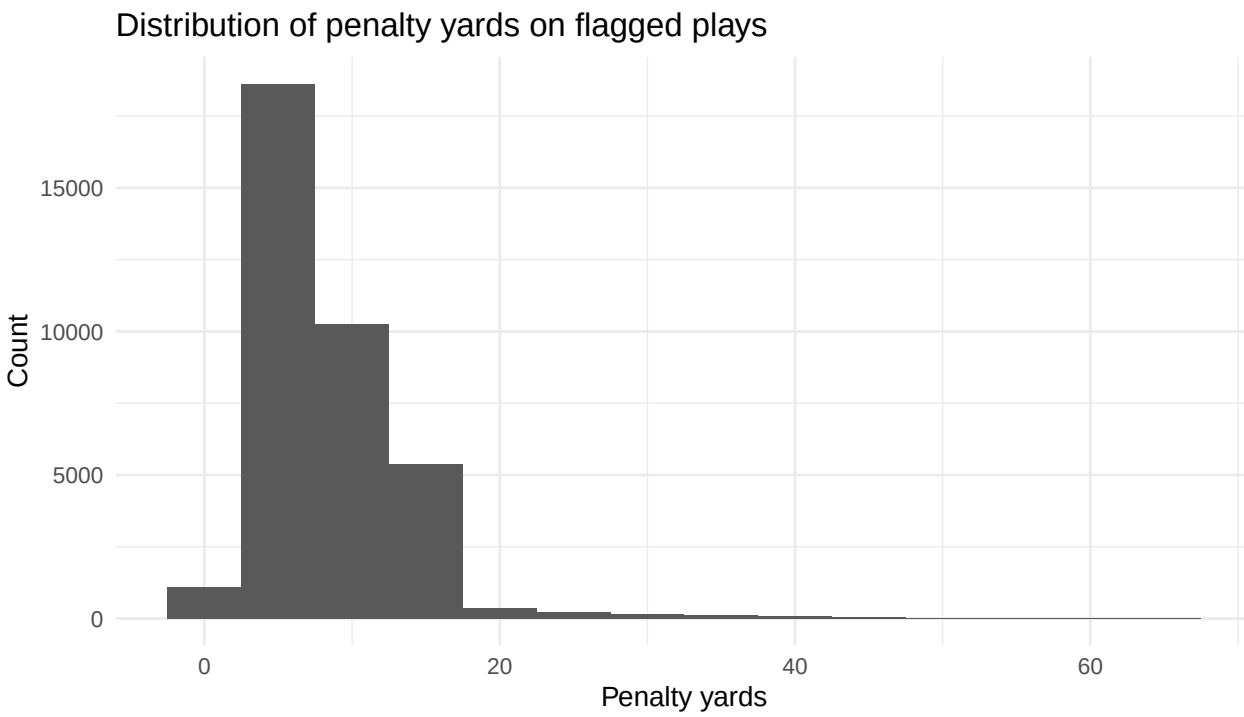
```r
penalty_by_season <- pbp %>%
  group_by(season) %>%
  summarise(
    plays = n(),
    penalties = sum(penalty == 1, na.rm = TRUE),
    penalty_rate = penalties / plays,
    avg_penalty_yards = mean(penalty_yards[penalty == 1], na.rm = TRUE),
    .groups = "drop"
  )

penalty_by_season %>%
  mutate(
    penalty_rate = percent(penalty_rate, accuracy = 0.01),
    avg_penalty_yards = round(avg_penalty_yards, 2)
  ) %>%
  gt()
```

```r
ggplot(penalty_by_season, aes(x = season, y = penalty_rate)) +
  geom_line() +
  geom_point() +
  scale_y_continuous(labels = percent_format(accuracy = 0.1)) +
  ylim(0.05, 0.08) +
  labs(
    title = "Penalty play rate by season",
    x = "Season",
    y = "Penalty plays / all play-by-play rows"
  ) +
  theme_minimal()
```



Distribution of penalty yards

```
penalty_plays %>%
  filter(!is.na(penalty_yards), penalty_yards >= 0) %>%
  ggplot(aes(x = penalty_yards)) +
  geom_histogram(binwidth = 5) +
  labs(
    title = "Distribution of penalty yards on flagged plays",
    x = "Penalty yards",
    y = "Count"
  ) +
  theme_minimal()
```



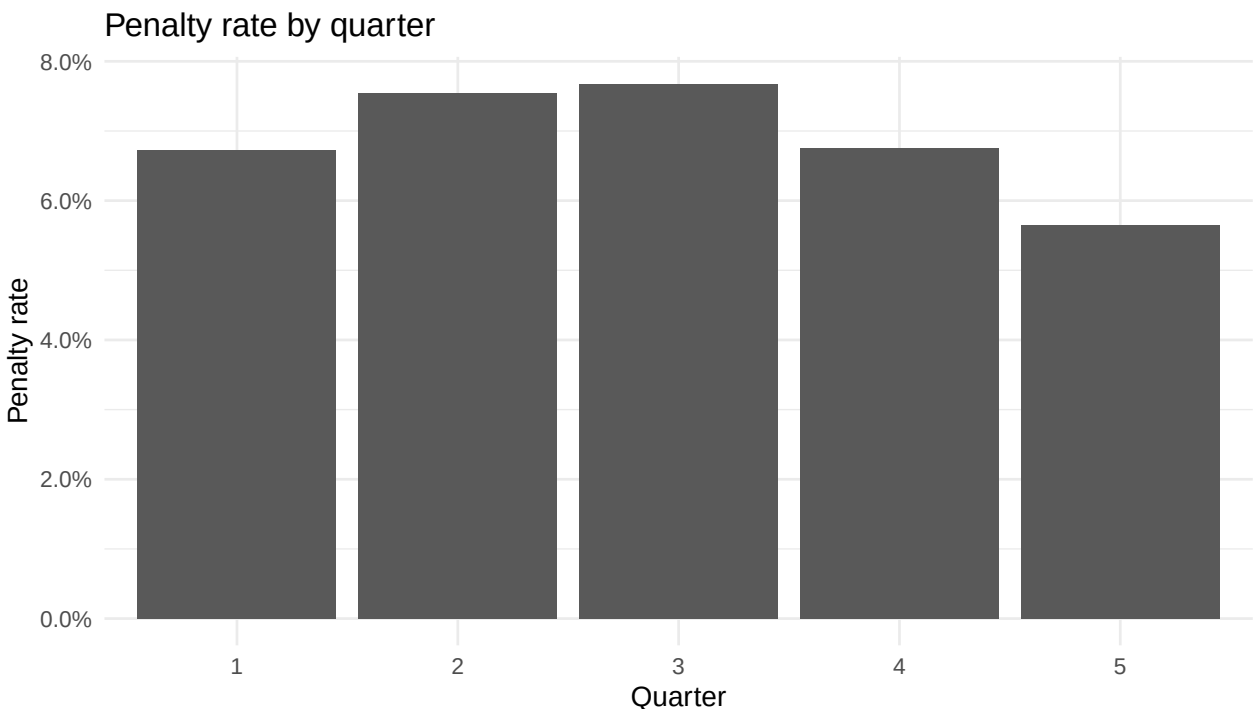
Which situations produce more penalties?

```
penalty_context <- pbp %>%
  mutate(
    is_penalty = penalty == 1,
    qtr = factor(qtr),
    down = factor(down),
    play_type = fct_lump_n(as.factor(play_type), n = 8)
  )

context_summary <- penalty_context %>%
  group_by(qtr, down, play_type) %>%
  summarise(
    plays = n(),
    penalties = sum(is_penalty, na.rm = TRUE),
    penalty_rate = penalties / plays,
    .groups = "drop"
  ) %>%
  filter(plays >= 100)

context_summary %>%
  arrange(desc(penalty_rate)) %>%
  slice_head(n = 10) %>%
  gt()
```

```
pbp %>%
  mutate(qtr = factor(qtr)) %>%
```

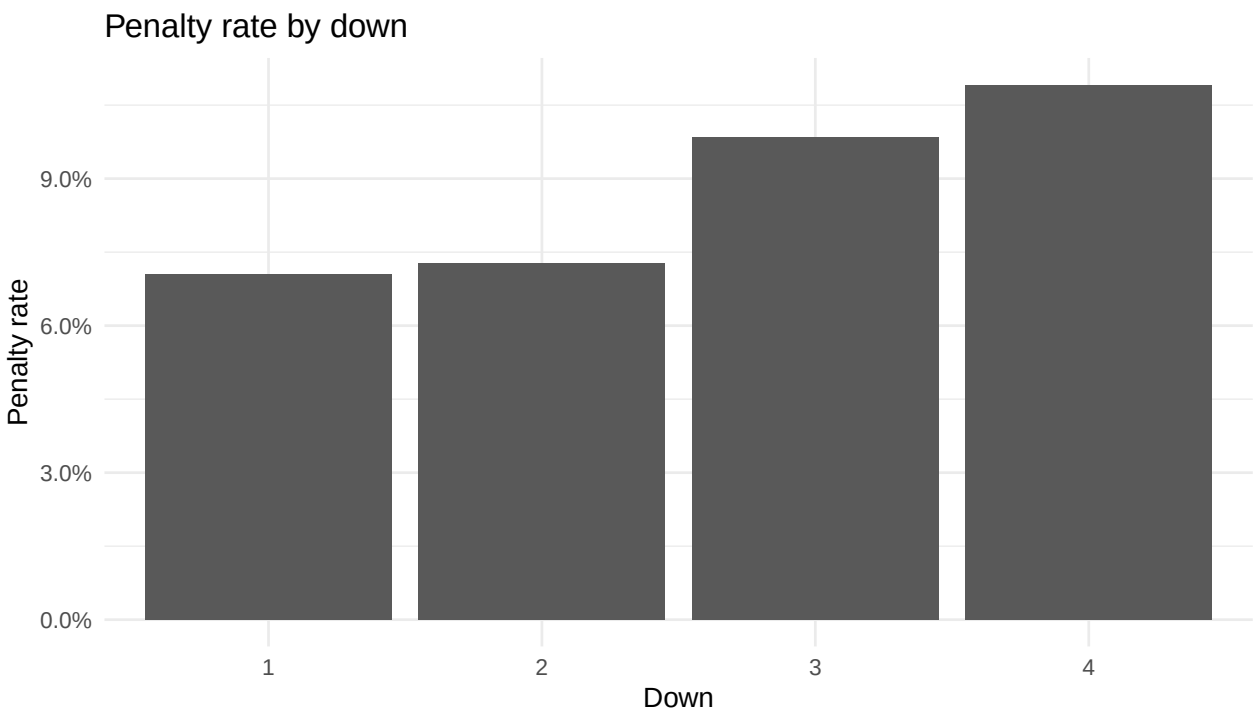
qtr	down	play_type	plays	penalties	penalty_rate
4	1	no_play	2387	2353	0.9857562
2	1	no_play	2965	2917	0.9838111
1	1	no_play	2094	2060	0.9837631
4	2	no_play	1954	1920	0.9825998
1	3	no_play	1601	1573	0.9825109
2	2	no_play	2335	2292	0.9815846
3	3	no_play	1546	1517	0.9812419
3	2	no_play	1807	1773	0.9811843
1	2	no_play	1582	1552	0.9810367
3	1	no_play	2335	2289	0.9802998

```r
  group_by(qtr) %>%
  summarise(
    plays = n(),
    penalties = sum(penalty == 1, na.rm = TRUE),
    rate = penalties / plays,
    .groups = "drop"
  ) %>%
  ggplot(aes(x = qtr, y = rate)) +
  geom_col() +
  scale_y_continuous(labels = percent_format(accuracy = 0.1)) +
  labs(
    title = "Penalty rate by quarter",
    x = "Quarter",
    y = "Penalty rate"
  ) +
  theme_minimal()
```




```
pbp %>%
  filter(!is.na(down)) %>%
  mutate(down = factor(down)) %>%
  group_by(down) %>%
  summarise(
    plays = n(),
    penalties = sum(penalty == 1, na.rm = TRUE),
    rate = penalties / plays,
    .groups = "drop"
  ) %>%
  ggplot(aes(x = down, y = rate)) +
  geom_col() +
  scale_y_continuous(labels = percent_format(accuracy = 0.1)) +
  labs(
    title = "Penalty rate by down",
    x = "Down",
    y = "Penalty rate"
  ) +
  theme_minimal()
```

penalty_team	n
LV	1251
DAL	1230
SEA	1217
CLE	1215
BUF	1208
JAX	1207
NYJ	1203
MIA	1196
DEN	1187
TB	1187
BAL	1184
KC	1183
ARI	1180
NO	1175
HOU	1167



Most common penalty teams and penalty types

```
penalty_plays %>%
  count(penalty_team, sort = TRUE) %>%
  slice_head(n = 15) %>%
  gt()
```

```r
penalty_plays %>%
  select(c("game_id", "season", "week", "posteam", "defteam", "play_type", "desc", "penalty_yar
  slice_head(n = 5)
```

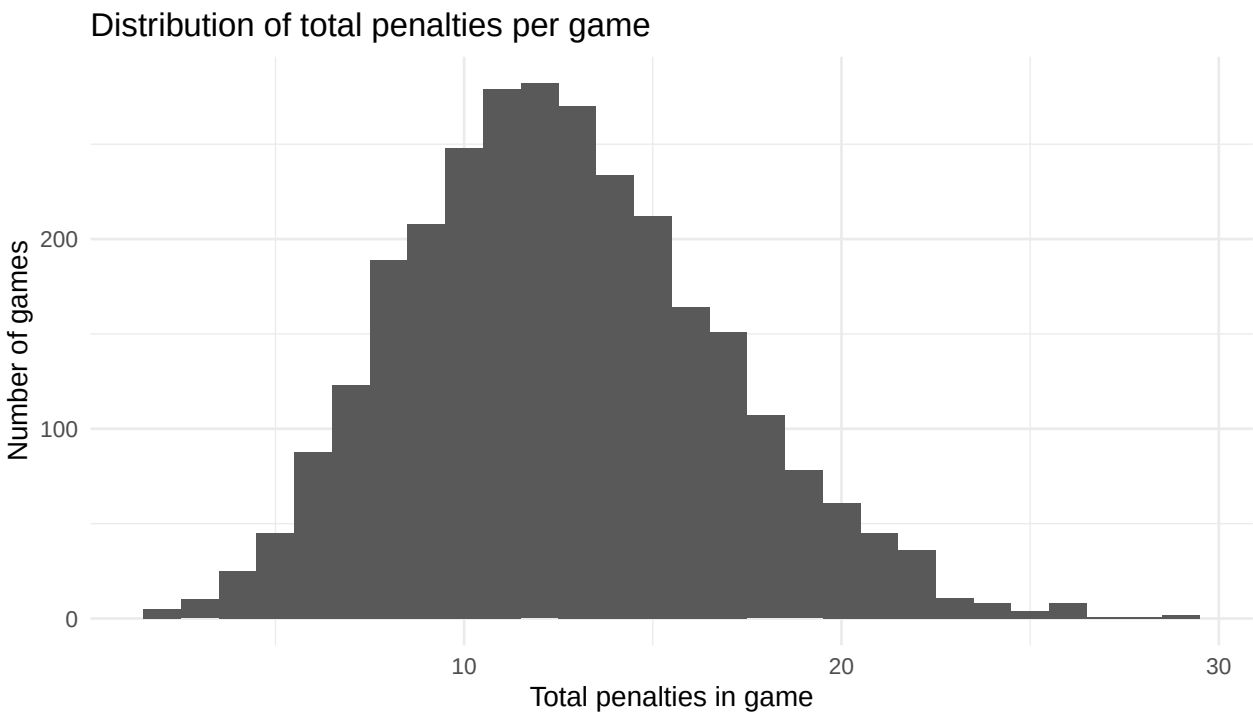
```
## # A tibble: 5 x 8
##   game_id         season  week posteam defteam play_type desc      penalty_yards
##   <chr>            <int> <int> <chr>   <chr>   <chr>     <chr>             <dbl>
## 1 2015_01_BAL_DEN   2015     1 DEN     BAL     no_play   (9:29) (~             5
## 2 2015_01_BAL_DEN   2015     1 DEN     BAL     no_play   (5:39) (~             5
## 3 2015_01_BAL_DEN   2015     1 DEN     BAL     no_play   (13:33) ~             5
## 4 2015_01_BAL_DEN   2015     1 DEN     BAL     no_play   (12:31) ~             5
## 5 2015_01_BAL_DEN   2015     1 DEN     BAL     no_play   (3:38) (~             5
```

Game-level aggregation from play-by-play

```r
game_level <- pbp %>%
  group_by(game_id, old_game_id, season, week, season_type, home_team, away_team) %>%
  summarise(
    total_rows = n(),
    total_penalties = sum(penalty == 1, na.rm = TRUE),
    total_penalty_yards = sum(penalty_yards, na.rm = TRUE),
    unique_posteams = n_distinct(posteam, na.rm = TRUE),
    mean_abs_score_diff = mean(abs(score_differential), na.rm = TRUE),
    mean_ydstogo = mean(ydstogo, na.rm = TRUE),
    pass_rate = mean(play_type == "pass", na.rm = TRUE),
    rush_rate = mean(play_type == "run", na.rm = TRUE),
    .groups = "drop"
  )
```

```r
ggplot(game_level, aes(x = total_penalties)) +
  geom_histogram(binwidth = 1) +
  labs(
    title = "Distribution of total penalties per game",
    x = "Total penalties in game",
    y = "Number of games"
  ) +
  theme_minimal()
```

mean_penalties	var_penalties	overdispersion_ratio
12.59378	17.49353	1.389061




```
game_level %>%
  summarise(
    mean_penalties = mean(total_penalties, na.rm = TRUE),
    var_penalties = var(total_penalties, na.rm = TRUE),
    overdispersion_ratio = var_penalties / mean_penalties
  ) %>%
  gt()
```

Team-game aggregation from play-by-play

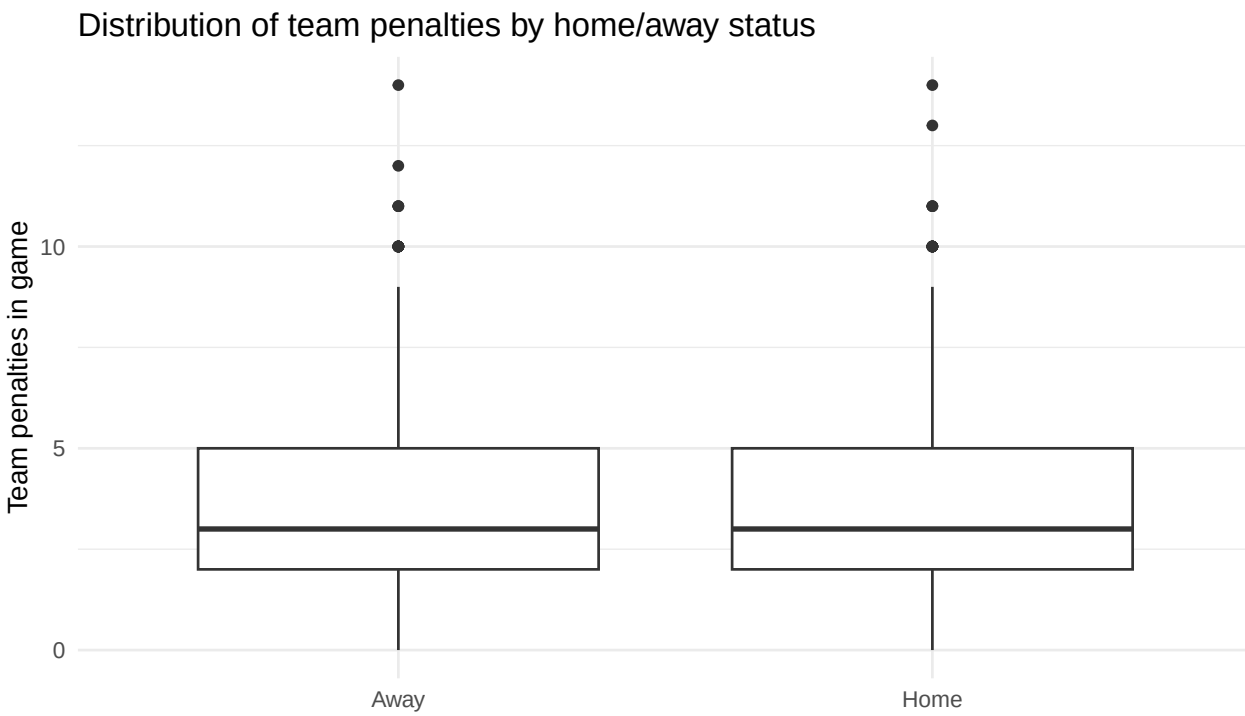
```
team_game_level <- pbp %>%
  filter(!is.na(posteam)) %>%
  group_by(game_id, old_game_id, season, week, season_type, home_team, away_team, posteam) %>%
  summarise(
    team_plays = n(),
    team_penalties = sum(penalty == 1 & penalty_team == first(posteam), na.rm = TRUE),
    team_penalty_yards = sum(if_else(penalty_team == first(posteam), penalty_yards, 0), na.rm =
    offensive_penalty_rows = sum(penalty == 1 & posteam == penalty_team, na.rm = TRUE),
    mean_ydstogo = mean(ydstogo, na.rm = TRUE),
    mean_abs_score_diff = mean(abs(score_differential), na.rm = TRUE),
    early_down_rate = mean(down %in% c(1, 2), na.rm = TRUE),
    pass_rate = mean(play_type == "pass", na.rm = TRUE),
    rush_rate = mean(play_type == "run", na.rm = TRUE),
```

home_away	mean_penalties	median_penalties	mean_penalty_yards
Away	3.451813	3	26.36408
Home	3.407599	3	26.19344

```
    is_home = as.integer(first(posteam) == first(home_team)),
    .groups = "drop"
  )
```

```
team_game_level %>%
  mutate(home_away = if_else(is_home == 1, "Home", "Away")) %>%
  group_by(home_away) %>%
  summarise(
    mean_penalties = mean(team_penalties, na.rm = TRUE),
    median_penalties = median(team_penalties, na.rm = TRUE),
    mean_penalty_yards = mean(team_penalty_yards, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  gt()
```

```
team_game_level %>%
  mutate(home_away = if_else(is_home == 1, "Home", "Away")) %>%
  ggplot(aes(x = home_away, y = team_penalties)) +
  geom_boxplot() +
  labs(
    title = "Distribution of team penalties by home/away status",
    x = NULL,
    y = "Team penalties in game"
  ) +
  theme_minimal()
```



3) EDA: Officials data

```
officials <- officials_raw %>%
  filter(season %in% seasons) %>%
  filter(if ("season_type" %in% names(.)) season_type %in% season_type_keep else TRUE)

officials %>%
  summarise(
    rows = n(),
    games = n_distinct(game_id),
    officials = n_distinct(official_name),
    positions = n_distinct(position)
  )
```

```
## # A tibble: 1 x 4
##    rows games officials positions
##   <int> <int>     <int>     <int>
## 1 20464  2896       223        11
```

```
officials %>%
  slice_head(n = 5)
```

```
## # A tibble: 5 x 9
##   game_id    game_key official_name  position   jersey_number official_id season
##   <chr>      <chr>    <chr>          <chr>              <int> <chr>        <int>
## 1 2015091000 56503    Brad Freeman   Field Jud~            88 25            2015
## 2 2015091000 56503    Kent Payne     Head Line~            79 28            2015
```

lead_referee	n
Craig Wrolstad	172
Bill Vinovich	171
Clete Blakeman	171
Ronald Torbert	171
Carl Cheffers	170
Brad Allen	169
John Hussey	169
Alex Kemp	126
Shawn Smith	126
Shawn Hochuli	124

```
## 3 2015091000 56503    Terrence Miles Back Judge           111 139           2015
## 4 2015091000 56503    Tim Podraza    Line Judge            47 123           2015
## 5 2015091000 56503    Scott Novak    Side Judge             1 94            2015
## # i 2 more variables: season_type <chr>, week <int>
```

Extract lead referee summaries

```
referee_game <- officials %>%
  filter(position == "Referee") %>%
  transmute(
    game_id,
    season,
    week,
    lead_referee = official_name,
    lead_referee_id = official_id,
    referee_jersey = jersey_number
  )

referee_game %>%
  summarise(games = n(), unique_referees = n_distinct(lead_referee))
```

```
## # A tibble: 1 x 2
##   games unique_referees
##   <int>           <int>
## 1  2896              29
```

```
referee_game %>%
  count(lead_referee, sort = TRUE) %>%
  slice_head(n = 10) %>%
  gt()
```

4) Joined EDA: penalties merged with officiating data

```
referee_game <- officials %>%
  filter(position == "Referee") %>%
  transmute(
    old_game_id = game_id,
    season,
    week,
    lead_referee = official_name,
    lead_ref_id = official_id
  )
```

```
game_joined <- game_level %>%
  left_join(referee_game, by = c("old_game_id", "season", "week"))

game_joined %>%
  summarise(
    games = n(),
    missing_referee = sum(is.na(lead_referee)),
    pct_missing_referee = mean(is.na(lead_referee))
  )
```

```
## # A tibble: 1 x 3
##   games missing_referee pct_missing_referee
##   <int>           <int>               <dbl>
## 1  2895               3             0.00104
```

```
game_joined %>%
  select(c("game_id", "season", "week", "home_team", "away_team",
               "total_penalties", "total_penalty_yards", "total_rows",
               "lead_referee")) %>%
  slice_head(n = 5)
```

```
## # A tibble: 5 x 9
##   game_id   season  week home_team away_team total_penalties total_penalty_yards
##   <chr>      <int> <int> <chr>     <chr>               <int>               <dbl>
## 1 2015_01_~   2015     1 DEN       BAL                    11                  60
## 2 2015_01_~   2015     1 JAX       CAR                     6                  47
## 3 2015_01_~   2015     1 LV        CIN                    11                  82
## 4 2015_01_~   2015     1 NYJ       CLE                    16                 139
## 5 2015_01_~   2015     1 LAC       DET                     9                  69
## # i 2 more variables: total_rows <int>, lead_referee <chr>
```

Team-game joined dataset

```
team_game_joined <- team_game_level %>%
  left_join(referee_game, by = c("old_game_id", "season", "week"))

table(
```

```
  has_referee = !is.na(team_game_joined$lead_referee),
  useNA = "ifany"
)
```

```
## has_referee
## FALSE  TRUE
##     6  5784
```

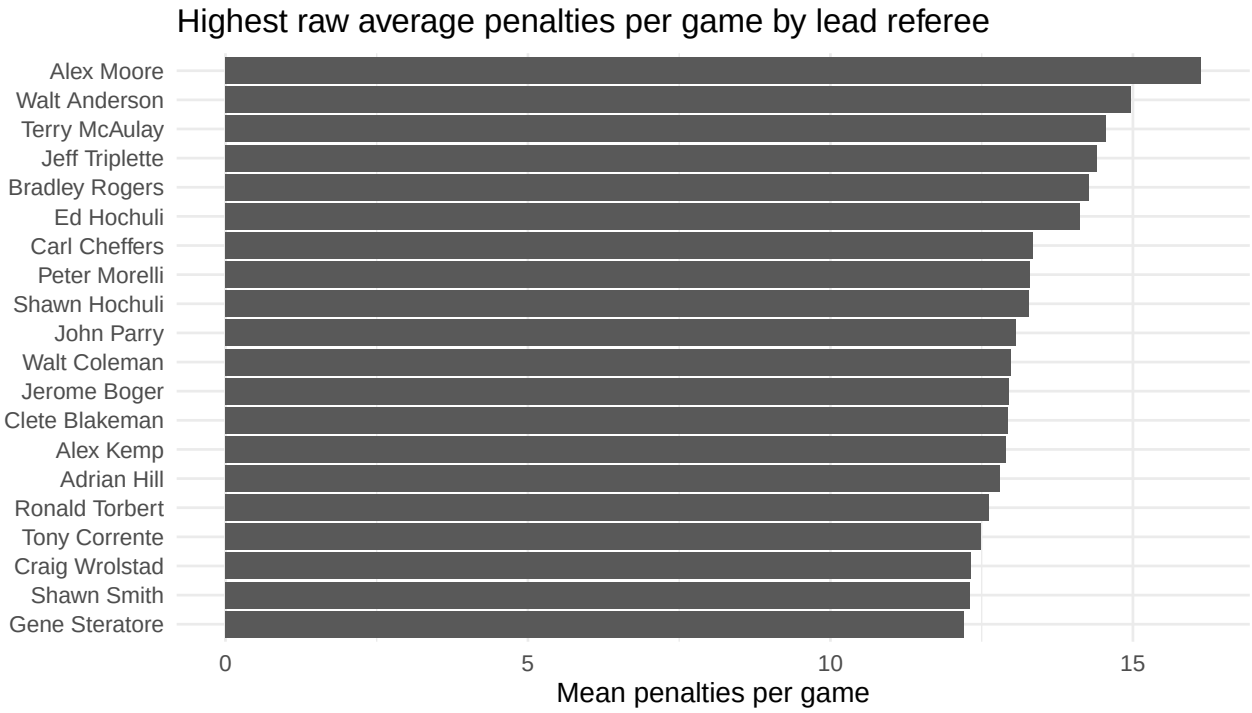
Raw referee differences: do some refs call more penalties?

Referee-level game averages

```
ref_game_summary <- game_joined %>%
  filter(!is.na(lead_referee)) %>%
  group_by(lead_referee) %>%
  summarise(
    games = n(),
    mean_penalties = mean(total_penalties, na.rm = TRUE),
    median_penalties = median(total_penalties, na.rm = TRUE),
    sd_penalties = sd(total_penalties, na.rm = TRUE),
    mean_penalty_yards = mean(total_penalty_yards, na.rm = TRUE),
    mean_total_rows = mean(total_rows, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  filter(games >= 8) %>%
  arrange(desc(mean_penalties))

ref_game_summary %>%
  slice_head(n = 20) %>%
  gt()
```

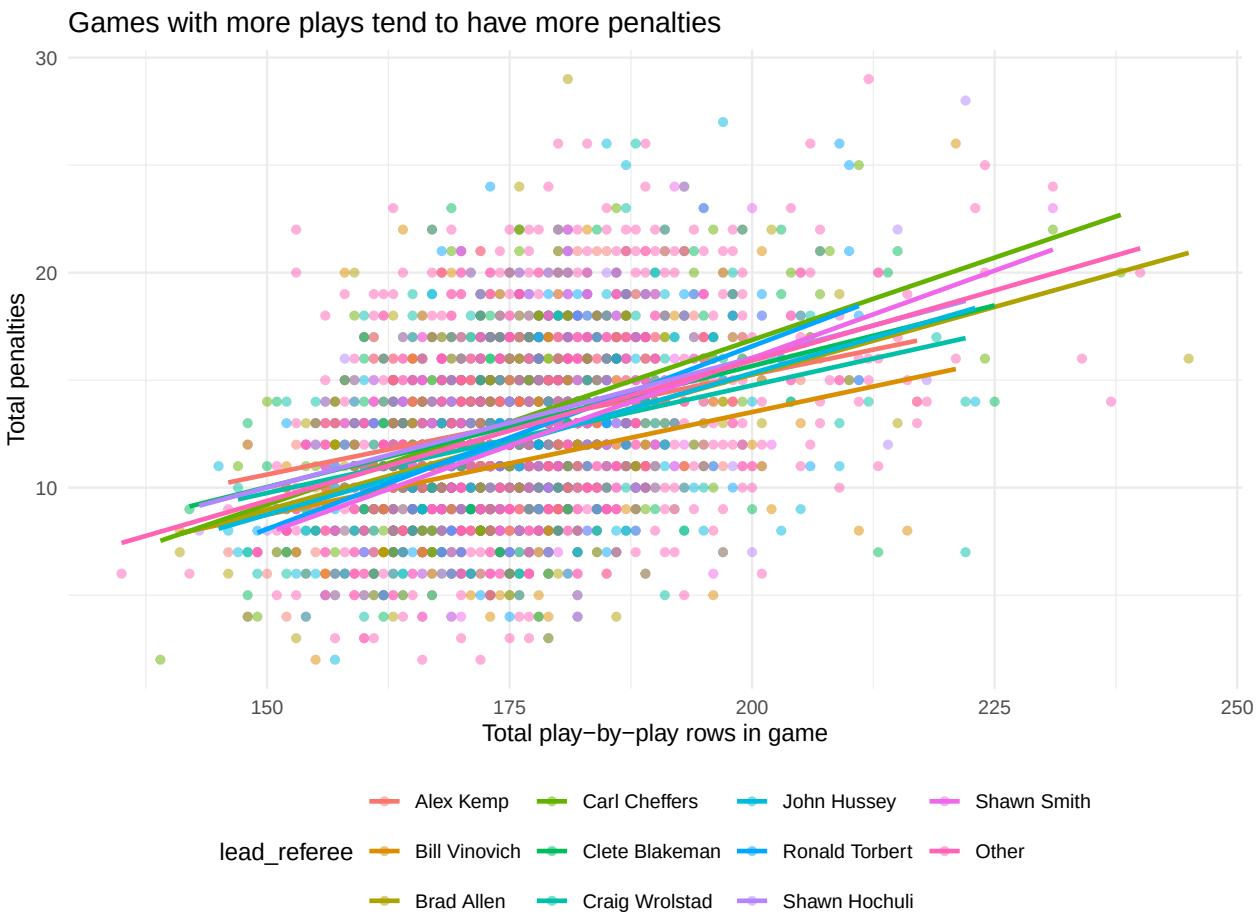
```
ref_game_summary %>%
  slice_max(order_by = mean_penalties, n = 20) %>%
  mutate(lead_referee = fct_reorder(lead_referee, mean_penalties)) %>%
  ggplot(aes(x = lead_referee, y = mean_penalties)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Highest raw average penalties per game by lead referee",
    x = NULL,
    y = "Mean penalties per game"
  ) +
  theme_minimal()
```

lead_referee	games	mean_penalties	median_penalties	sd_penalties	mean_penalty_yards	m
Alex Moore	16	16.12500	17	4.558874	127.1875	
Walt Anderson	76	14.96053	15	4.116441	129.4474	
Terry McAulay	43	14.55814	14	4.090071	125.1628	
Jeff Triplette	45	14.40000	14	4.783494	126.2667	
Bradley Rogers	15	14.26667	14	4.399134	123.1333	
Ed Hochuli	46	14.13043	14	3.303928	121.8043	
Carl Cheffers	170	13.34118	14	4.274745	113.1529	
Peter Morelli	60	13.30000	13	4.847855	117.1167	
Shawn Hochuli	124	13.28226	13	3.954128	113.6855	
John Parry	60	13.06667	13	4.513132	112.8167	
Walt Coleman	62	12.98387	12	4.166580	108.6774	
Jerome Boger	123	12.95122	12	4.337888	109.3171	
Clete Blakeman	171	12.93567	13	4.178582	106.8070	
Alex Kemp	125	12.89600	13	3.715321	109.9680	
Adrian Hill	109	12.80734	13	3.994156	107.4128	
Ronald Torbert	171	12.61404	12	4.378105	107.4561	
Tony Corrente	107	12.48598	12	4.525326	104.5514	
Craig Wrolstad	172	12.31977	12	4.090649	105.6628	
Shawn Smith	125	12.30400	12	3.890070	104.9840	
Gene Steratore	45	12.20000	12	3.584056	100.4889	



Relationship between game pace and penalties, colored by referee

```
set.seed(123)
plot_data <- game_joined %>%
  filter(!is.na(lead_referee)) %>%
  mutate(lead_referee = fct_lump_n(lead_referee, n = 10))

ggplot(plot_data, aes(x = total_rows, y = total_penalties, color = lead_referee)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Games with more plays tend to have more penalties",
    x = "Total play-by-play rows in game",
    y = "Total penalties"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```



Home / away penalty differences under different referees

```
ref_home_away <- team_game_joined %>%
  filter(!is.na(lead_referee)) %>%
```

lead_referee	home_away	games	mean_team_penalties
Adrian Hill	Away	109	3.779817
Adrian Hill	Home	109	3.284404
Alan Eck	Away	48	3.187500
Alan Eck	Home	48	3.270833
Alex Kemp	Away	125	3.464000
Alex Kemp	Home	125	3.416000
Alex Moore	Away	16	4.125000
Alex Moore	Home	16	4.312500
Bill Vinovich	Away	171	2.929825
Bill Vinovich	Home	171	2.859649

```
  mutate(home_away = if_else(is_home == 1, "Home", "Away")) %>%
  group_by(lead_referee, home_away) %>%
  summarise(
    games = n(),
    mean_team_penalties = mean(team_penalties, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  filter(games >= 8)

ref_home_away %>%
  slice_head(n = 10) %>%
  gt()
```

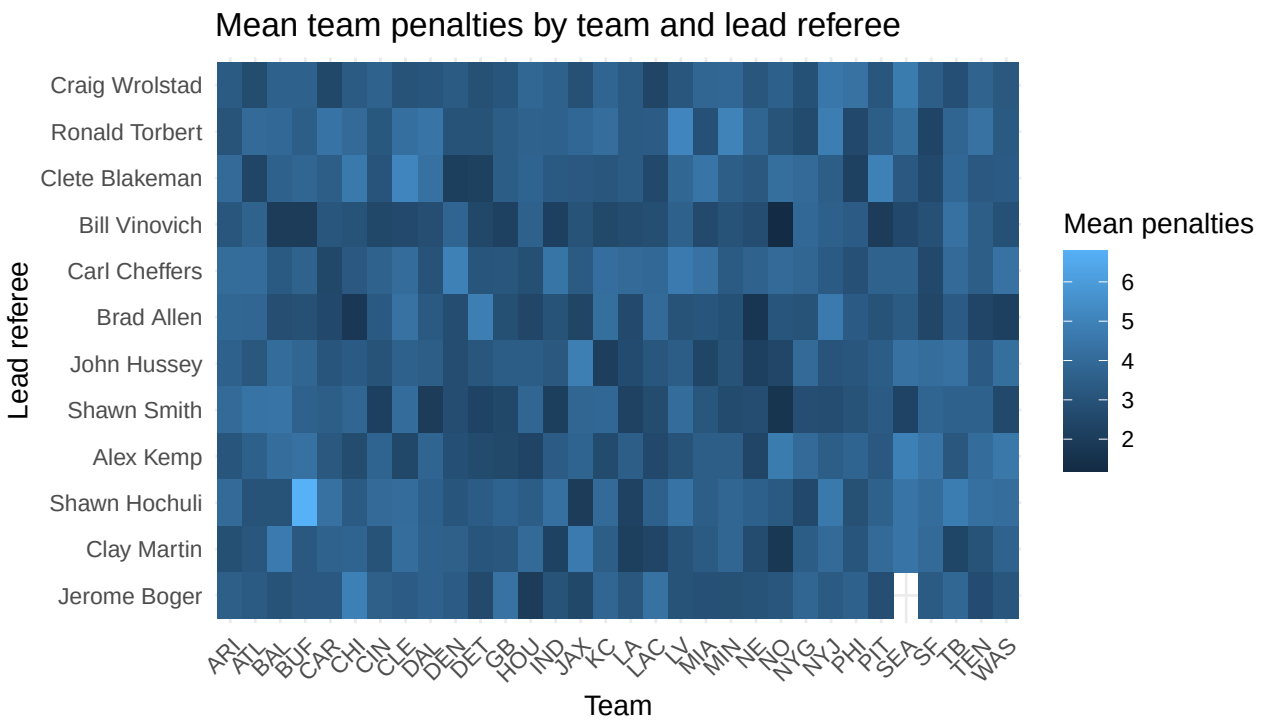
Team and referee interaction patterns

```
team_ref_summary <- team_game_joined %>%
  filter(!is.na(lead_referee)) %>%
  group_by(lead_referee, posteam) %>%
  summarise(
    games = n(),
    mean_penalties = mean(team_penalties, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  filter(games >= 3)

team_ref_summary %>%
  arrange(desc(games)) %>%
  slice_head(n = 10) %>%
  gt()
```

```
heatmap_data <- team_ref_summary %>%
  group_by(lead_referee) %>%
  summarise(total_games = sum(games), .groups = "drop") %>%
```

lead_referee	posteam	games	mean_penalties
Carl Cheffers	KC	17	4.176471
Carl Cheffers	SEA	17	3.705882
Clete Blakeman	TEN	16	3.250000
John Hussey	LA	16	2.687500
Bill Vinovich	DAL	15	2.800000
Brad Allen	BAL	15	2.800000
Brad Allen	CLE	15	4.333333
Carl Cheffers	ARI	15	4.066667
Craig Wrolstad	IND	15	3.600000
Bill Vinovich	MIN	14	3.000000

```
  slice_max(order_by = total_games, n = 12) %>%
  select(lead_referee) %>%
  inner_join(team_ref_summary, by = "lead_referee")

ggplot(heatmap_data, aes(x = posteam, y = fct_reorder(lead_referee, games, .fun = sum), fill =
  geom_tile() +
  labs(
    title = "Mean team penalties by team and lead referee",
    x = "Team",
    y = "Lead referee",
    fill = "Mean penalties"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```


Mean team penalties by team and lead referee
Lead referee
Craig Wrolstad
Ronald Torbert
Clete Blakeman
Bill Vinovich
Carl Cheffers
Brad Allen
John Hussey
Shawn Smith
Alex Kemp
Shawn Hochuli
Clay Martin
Jerome Boger
ARI
ATL
BAL
BUF
CAR
CHI
CIN
CLE
DAL
DEN
DET
GB
HOU
IND
JAX
KC
LA
LAC
LV
MIA
MIN
NE
NO
NYG
NYJ
PHI
PIT
SEA
SF
TB
TEN
WAS
Team
Mean penalties
6
5
4
3
2